

Hierarchical Explanation Graph: A Causal Framework for Explainability in Hierarchical LLM-Based Multi-Agent Systems

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Abstract

Large language model (LLM) agents increasingly operate in hierarchical multi-agent pipelines, where decisions cascade across layers of specialized components — planners, memory systems, executors, and evaluators. When queried about their decisions, these agents produce natural language self-explanations that are plausible and confident, yet may be causally disconnected from the actual decision process — a phenomenon structurally analogous to human confabulation. Existing explainability methods do not generalize to hierarchical agent architectures, where causal influence is distributed across layers rather than localized to a single inference call.

I introduce the **Hierarchical Explanation Graph (HEG)**, a two-level explainability framework that distinguishes between *intentional explanations* — what each layer reports about its own decision — and *causal explanations* — what controlled perturbation experiments empirically show drove the decision. A dual-orchestrator architecture enables real-time causal intervention: Orchestrator A executes the normal pipeline and logs all true intermediate states along with layer-level self-explanations, while Orchestrator B perturbs each intermediate state using a layer-aware Perturbation Engine and cascades the intervention downstream, yielding final outputs from which causal weights are computed via cosine divergence.

I evaluate the HEG framework across 7 tasks of varying complexity spanning a 6-layer pipeline (Global Planner, Mid-Level Planner, Memory System, Research Agent, Analysis Agent, Critic Evaluator), conducting 42 total perturbation experiments. Results reveal systematic intentional/causal discrepancy: the Critic Evaluator exhibits a 100% confabulation rate with near-zero causal weight, the Analysis Agent 43%, while execution-level layers (Research Agent, Mid-Level Planner) show zero confabulation. Causal weight distributions shift significantly between simple and multi-step tasks, providing empirical evidence that pipeline transparency requires task-aware, layer-resolved causal attribution rather than post-hoc narrative evaluation.

1 Introduction

The deployment of large language model agents capable of planning, reasoning, and executing complex multi-step tasks has introduced a new class of explainability challenges. Unlike single-model inference, hierarchical multi-agent systems route user goals through a cascade of specialized components — strategic planners decompose objectives, memory systems retrieve contextual knowledge, specialized agents collect and interpret data, and critic modules evaluate the final output. Each layer in this pipeline makes decisions that constrain and inform downstream behavior, creating a distributed causal structure that is opaque at every level.

A naive response to this opacity is to ask the agent to explain itself. Modern LLM agents can produce fluent, detailed natural language explanations of their decisions: “I selected Plan B because it maximized task completeness given the user’s time constraint.” These explanations are syntactically coherent and semantically plausible. However, they are generated by the same probabilistic mechanism that produced the decision — they are not verified against the actual causal process. An agent that retrieves a specific memory chunk may attribute its decision to an explicit goal constraint while the memory chunk was the true causal driver; the self-report passes no internal consistency check against the computational reality.

This failure mode is structurally identical to *confabulation* in cognitive science. First documented in patients with Korsakoff syndrome [Dalla Barba, 1993], confabulation refers to the production of fabricated memories or explanations without conscious intent to deceive. Critically, confabulated explanations pass the confabulator’s own consistency checks — the person genuinely believes the account they provide. Psychological research has long established that verbal self-reports of decision causes are unreliable [Wegner, 2002], and that distinguishing faithful from plausible explanations requires external intervention [Johnson, 1991]. I argue that the same epistemological problem applies to LLM agent self-explanation, and that addressing it requires the same solution: controlled causal intervention rather than self-report alone.

Existing explainability methods fail to address this in the multi-agent setting. Gradient-based attribution methods such as SHAP [Lundberg and Lee, 2017] and Grad-CAM [Selvaraju et al., 2017] operate at the token or feature level of a single model and do not generalize to sequential multi-layer pipelines. Chain-of-thought prompting [Wei et al., 2022] improves intermediate reasoning transparency within a single call but provides no mechanism to verify whether the stated reasoning reflects the actual causal computation. Agent transparency frameworks [Yao et al., 2023, Shinn et al., 2023] expose action traces without distinguishing the causal from the intentional level. To my knowledge, no prior framework proposes real-time causal intervention as an explainability instrument for hierarchical multi-agent systems.

I introduce the **Hierarchical Explanation Graph (HEG)**, a two-level causal explainability framework grounded in a cognitive science analogy. Just as psychologists cannot directly observe human intentions but must infer them from behavior, context, and — with appropriate skepticism — verbal reports [Dennett, 1987], I cannot directly observe the causal mechanism of a hierarchical agent’s decision. I can only observe its outputs and its self-reports. The HEG framework operationalizes this epistemological position by combining simultaneous intentional explanation capture at each pipeline layer with a systematic program of layer-targeted causal interventions.

The main contributions of this work are as follows:

- A **dual-orchestrator architecture** for real-time causal intervention on live hierarchical LLM pipelines, enabling layer-specific perturbation experiments without exponential complexity.
- A **layer-aware Perturbation Engine** that generates minimal, semantically meaningful interventions tailored to the output type of each pipeline layer.
- A **causal weight computation method** based on cosine divergence between original and perturbed final outputs, producing a normalized causal attribution distribution over all pipeline layers.
- The **Hierarchical Explanation Graph**, a weighted directed graph that overlays experimentally grounded causal weights onto the agent architecture, making the causal chain visually and quantitatively interpretable.
- **Empirical evidence** across 7 tasks and 42 perturbation experiments demonstrating systematic intentional/causal discrepancy, with the Critic Evaluator exhibiting a 100% confabulation rate despite being the layer that produces the final answer.

The remainder of this paper is organized as follows. Section 2 reviews related work in explainability, causal inference, LLM agent transparency, and cognitive science confabulation theory. Section 3 presents the HEG framework in full, covering the theoretical grounding, formal definitions, dual-orchestrator implementation, Perturbation Engine design, and causal weight computation. Section 4 reports experimental results across 7 tasks. Section 5 discusses the implications of my findings. Section 6 concludes and outlines future directions.

2 Related Work

Explainability in Neural Networks. The field of explainable AI has produced a rich set of post-hoc attribution methods for neural networks. LIME [Ribeiro et al., 2016] approximates model behavior locally using interpretable surrogates. SHAP [Lundberg and Lee, 2017] assigns feature importances grounded in Shapley values from cooperative game theory. Grad-CAM [Selvaraju et al., 2017] localizes attention in convolutional networks using gradient signals. While these methods have been influential, they share a fundamental limitation: they operate on a single forward pass of a single model. In hierarchical multi-agent pipelines, the unit of computation is not a forward pass but a multi-layer cascade of interacting LLM calls, and causal influence is distributed across this cascade rather than attributable to input features.

Faithfulness and Interpretability in Language Models. Jacovi and Goldberg [2020] draw a critical distinction between *faithful* explanations — those that correctly reflect the model’s reasoning process — and *plausible* explanations — those that appear reasonable to a human. They argue that most introspective explanation methods produce the latter. Lipton [2018] similarly critiques the conflation of interpretability with post-hoc rationalization. Elhage et al. [2021] advance mechanistic interpretability by analyzing circuits in transformer models, providing a basis for causal understanding at the attention level. My work addresses the faithfulness problem at the agent system level, where the relevant computational units are layer outputs rather than attention heads. Doshi-Velez and Kim [2017] establish a rigorous framework for evaluating interpretability, emphasizing the need for task-grounded evaluation — a principle I operationalize through my 7-task empirical protocol.

LLM Agents and Chain-of-Thought. Wei et al. [2022] demonstrate that prompting LLMs to produce intermediate reasoning steps (chain-of-thought) improves performance on complex tasks and provides a window into the model’s reasoning. However, subsequent work [Jacovi and Goldberg, 2020] has shown that these intermediate steps are not necessarily faithful to the underlying computation. ReAct [Yao et al., 2023] interleaves reasoning and acting in agent settings, while Reflexion [Shinn et al., 2023] adds a self-reflection loop that enables agents to revise decisions based on prior feedback. Voyager [Wang et al., 2023] and Generative Agents [Park et al., 2023] demonstrate the power of persistent memory and planning in complex environments. None of these frameworks address the faithfulness of the explanations produced within the agent pipeline; they assume that the agent’s stated reasoning reflects its actual computation.

Causal Inference and Counterfactual Explanations. Pearl’s causal framework [Pearl, 2009] provides the theoretical foundation for understanding causal relationships through interventions (do-calculus). Counterfactual explanations [Wachter et al., 2017] ask: what is the smallest change to the input that would change the output? My perturbation methodology is in the spirit of counterfactual causal inference — I intervene on specific intermediate states and observe how the final output changes — but applied to intermediate pipeline states rather than input features, and across a hierarchical multi-layer architecture rather than a single model.

Confabulation and Cognitive Science. The term confabulation was introduced by Dalla Barba [1993] to describe a specific memory disorder in which patients produce fabricated accounts without conscious awareness of their falsity. Johnson [1991]’s reality monitoring framework explains this as a failure of the cognitive system that tags memories with their source — distinguishing self-generated from externally perceived content. Dennett [1987] introduced the intentional stance as a level of explanation at which we describe systems as having beliefs, desires, and intentions, independent of their underlying mechanism. Wegner [2002] demonstrated through a series of experiments that human explanations of voluntary action are often post-hoc rationalizations constructed after the fact. I argue that these cognitive science insights are directly applicable to LLM self-explanation: the agent’s verbal report of its reasoning is a post-hoc construction that may diverge from the actual causal mechanism, and detecting this divergence requires external intervention rather than self-report analysis alone.

3 Methodology

3.1 Theoretical Framework

The HEG framework is grounded in a cognitive science analogy that shapes both its architecture and its evaluation criteria. In psychology, the challenge of understanding another person’s intentions is structurally analogous to the challenge of understanding an agent’s decision. I never directly observe intentions — I infer them from behavior,

context, and verbal reports. Verbal self-reports are informative but unreliable: confabulation research demonstrates that people produce confident, coherent explanations of their own behavior that are causally disconnected from the actual mental process [Dalla Barba, 1993, Wegner, 2002]. To establish causal claims, psychologists perform interventions: manipulate a factor and observe whether the outcome changes [Pearl, 2009].

I apply this epistemological framework to hierarchical LLM agents. When layer l_i of a pipeline produces output O_i and simultaneously reports explanation X_i , the explanation is an intentional-level description that may or may not correspond to the actual causal factors. To determine the causal contribution of O_i to the final pipeline output, I perform a controlled intervention: I minimally perturb O_i and observe the change in the final output. This produces a causal weight W_i that is experimentally grounded rather than self-reported.

This leads to a two-level explainability framework. The *intentional level* answers: “What reason did the agent provide for its action?” It is valuable for human alignment and interpretability but not necessarily faithful. The *causal level* answers: “What factors actually caused the agent to behave this way?” It is experimentally verified and faithful by construction. The *intentional/causal discrepancy* — the gap between these two levels — is the primary target of the HEG framework. When a layer’s self-explanation emphasizes factors that the perturbation analysis shows were causally insignificant, I classify the explanation as confabulated.

3.2 Formal Definition of the HEG

Let the agent pipeline be a directed acyclic graph $\mathcal{P} = (L, E)$, where $L = \{l_0, l_1, l_2, l_3, l_4, l_5\}$ is the ordered set of pipeline layers and E encodes information flow between consecutive layers. The six layers correspond to: Global Planner (l_0), Mid-Level Planner (l_1), Memory System (l_2), Research Agent (l_3), Analysis Agent (l_4), and Critic Evaluator (l_5).

Each layer l_i produces an *intermediate state*:

$$S_i = (O_i, X_i) \quad (1)$$

where $O_i \in \mathcal{O}$ is the structured output (a typed JSON object specific to the layer’s role) and $X_i \in \mathcal{X}$ is the layer’s natural language intentional self-explanation, produced simultaneously with O_i in a single LLM call.

The **Hierarchical Explanation Graph** is a weighted directed graph:

$$\mathcal{G} = (V, E, W) \quad (2)$$

where:

- $V = \{\text{goal}\} \cup \{l_0, \dots, l_5\}$ is the node set, including the user goal as the source node;
- E is the directed edge set encoding the pipeline’s causal flow;
- $W : V \setminus \{\text{goal}\} \rightarrow [0, 1]$ is the causal weight function satisfying $\sum_{i=0}^5 W(l_i) = 1$.

The causal weight $W(l_i)$ quantifies the proportional contribution of layer l_i ’s actual output to the final pipeline decision, as determined by controlled perturbation experiments (Section 3.5).

3.3 Implementation: Dual-Orchestrator Architecture

The HEG framework is implemented through a dual-orchestrator architecture that separates normal pipeline execution from causal experimentation.

Orchestrator A — Normal Pipeline. Orchestrator A executes the full 6-layer pipeline sequentially. At each layer l_i , the LLM is prompted via a structured system prompt that requires a JSON response containing exactly two fields: `output` (the layer’s structured result) and `explanation` (a natural language self-explanation specifying which inputs influenced the decision and why). This simultaneous capture — explanation produced in the same generation as the output, not reconstructed afterward — minimizes the confabulation risk that would arise from a separate explanation call after the decision has been made. Orchestrator A logs all intermediate states $S_0, \dots, S_5$ and the final answer $A = O_5.\text{final_answer}$.

The six layers and their output schemas are:

- l_0 (*Global Planner*): strategic plan with goal, constraints, success criteria, and priorities.
- l_1 (*Mid-Level Planner*): subtask list with dependency graph, workflow order, and complexity estimate.
- l_2 (*Memory System*): retrieved memory chunks ranked by relevance score.
- l_3 (*Research Agent*): collected data items with sources, tools used, and data gaps.
- l_4 (*Analysis Agent*): insights, conclusions, confidence level, and recommended action.
- l_5 (*Critic Evaluator*): quality, completeness, and faithfulness scores; verdict; and final answer.

Orchestrator B — Perturbation Controller. Orchestrator B receives all intermediate states from Orchestrator A and executes one perturbation experiment per layer. For each state S_i , Orchestrator B:

1. Calls the Perturbation Engine to generate perturbed output O'_i ;
2. Constructs a modified pipeline state where O_i is replaced by O'_i while all upstream states $O_0, \dots, O_{i-1}$ remain unchanged;
3. Re-runs all downstream layers $l_{i+1}, \dots, l_5$ with the modified state, using the Perturbation Engine outputs as inputs;
4. Records the resulting final answer $B_i = O'_5.\text{final_answer}$.

This produces 6 perturbation experiments per task, executed sequentially. The cascade from perturbation point i involves re-running $5 - i$ downstream layers: perturbing l_0 triggers a full 5-layer re-run; perturbing l_5 produces the perturbed output directly. This linear complexity — $O(n)$ experiments for an n -layer pipeline — contrasts with the exponential branching ($O(2^n)$) that would result from perturbing all layers simultaneously at each step.

3.4 Perturbation Engine

The Perturbation Engine is a dedicated LLM component whose sole function is to generate controlled, minimal perturbations of layer outputs. Its design addresses two requirements: perturbations must be *semantically meaningful* (testing a real causal hypothesis, not introducing noise) and *minimal* (smallest possible change, so that experiments across layers remain comparable).

The engine is **layer-aware**: the perturbation instruction differs by layer type, targeting the most informative single-factor change for each component. The per-layer instructions are:

- l_0 (Global Planner): remove one constraint *or* change one success criterion *or* alter one strategic priority.
- l_1 (Mid-Level Planner): remove one subtask *or* reorder two adjacent workflow steps *or* change one agent assignment.
- l_2 (Memory System): remove the highest-relevance chunk *or* replace one chunk’s content with semantically different content.
- l_3 (Research Agent): remove one collected data item *or* downgrade the highest-rated source reliability to “low.”
- l_4 (Analysis Agent): remove one insight *or* flip one conclusion to its opposite *or* change confidence level.
- l_5 (Critic Evaluator): change verdict to “revise” *or* reduce quality score by 0.3 *or* add one critical issue.

The Perturbation Engine is called with temperature $T = 0.7$ to encourage varied but realistic interventions across repeated runs. OpenAI’s native `response_format: json_object` mode is used to guarantee structured output, with a 3-attempt fallback strategy for robustness.

3.5 Causal Weight Computation

For each perturbation experiment $i \in \{0, \dots, 5\}$, I compute the cosine divergence between the semantic embedding of the original final output A and the perturbed final output B_i :

$$D_i = 1 - \cos(\text{emb}(A), \text{emb}(B_i)) = 1 - \frac{\text{emb}(A) \cdot \text{emb}(B_i)}{\|\text{emb}(A)\| \|\text{emb}(B_i)\|} \quad (3)$$

where $\text{emb}(\cdot)$ denotes sentence-level embeddings computed using the `all-MiniLM-L6-v2` model [Reimers and Gurevych, 2019]. A larger D_i indicates that perturbing layer l_i ’s output caused a larger semantic shift in the final answer, implying greater causal influence of that layer.

Causal weights are obtained by normalizing across all experiments:

$$W_i = \frac{D_i}{\sum_{j=0}^5 D_j} \quad (4)$$

so that $\sum_{i=0}^5 W_i = 1$. A layer with $W_i > \frac{1}{6} \approx 0.167$ exerts above-uniform causal influence on the pipeline output. The uniform baseline $\frac{1}{6}$ represents the null hypothesis that all layers contribute equally to the final decision.

Intentional/causal discrepancy for layer l_i is then classified by comparing the layer’s self-explanation X_i against its measured causal weight W_i . A dedicated LLM analyzer assesses whether the factors cited in X_i are consistent with the magnitude and rank of W_i , producing a three-way classification: *faithful* (explanation aligns with causal evidence), *partial* (partial alignment), or *confabulated* (explanation claims causal agency inconsistent with the measured weight).

4 Results

Experimental Setup. I evaluate the HEG framework on 7 tasks spanning two complexity levels: 3 simple tasks (Task 1: Topic Summary, Task 2: Factual Q&A, Task 3: Email Drafting) and 4 multi-step tasks (Task 4: Research Project Plan, Task 5: System Architecture, Task 6: Literature Review, Task 7: Product Roadmap). All experiments use GPT-4o as the backbone LLM for all pipeline layers and the Perturbation Engine, sentence embeddings via `all-MiniLM-L6-v2`, and a 6-layer pipeline yielding $7 \times 6 = 42$ total perturbation experiments.

Finding 1: Causal weights are non-uniform and task-dependent. Figure 1 shows the causal weight per layer across all 7 tasks. No task exhibits a uniform causal distribution. The Global Planner dominates in simple tasks, reaching $W_0 = 0.570$ for Email Drafting — the highest single-layer causal weight observed across all experiments. The Critic Evaluator exhibits a near-zero causal weight ($W_5 \approx 0.000$) in every task without exception, a finding I examine in detail in Section 5.

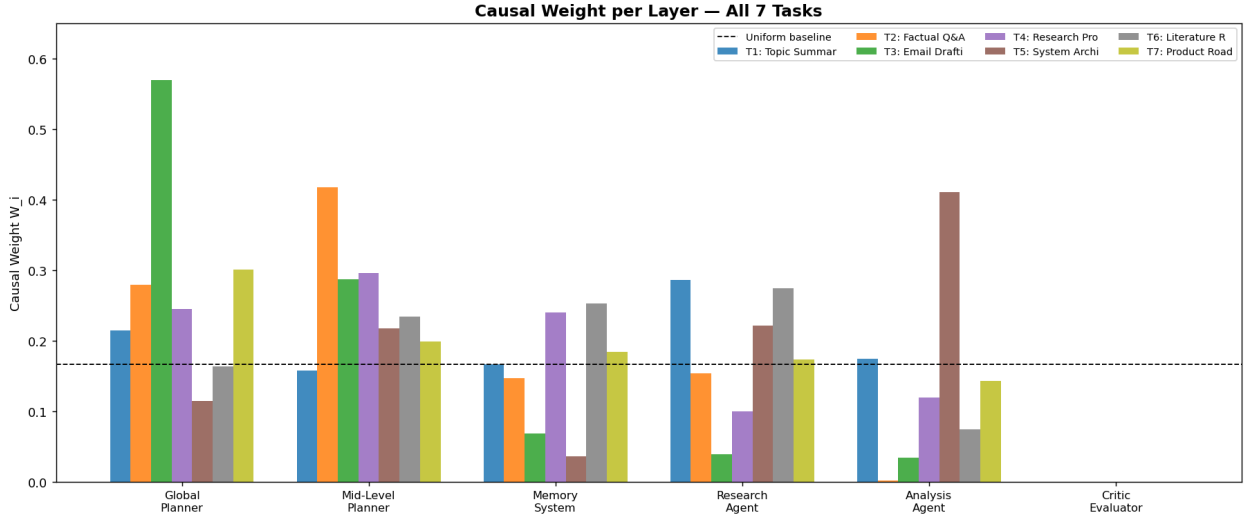


Figure 1: Causal weight W_i per layer across all 7 tasks. Each cluster of bars represents one pipeline layer; each color represents one task. The dashed line marks the uniform baseline $W = 1/6 \approx 0.167$. The Critic Evaluator (rightmost cluster) exhibits near-zero weight in all tasks, while the Global Planner and Mid-Level Planner dominate in simple tasks (T1–T3).

Finding 2: Simple and multi-step tasks exhibit distinct causal profiles. Figure 2 compares mean causal attribution per layer between simple and multi-step tasks. Simple tasks exhibit a strongly front-loaded profile: the Global Planner accounts for mean $W_0 \approx 0.355$ and the Mid-Level Planner for mean $W_1 \approx 0.290$, with all downstream layers below baseline. Multi-step tasks show a substantially more uniform distribution — all layers from Global Planner through Analysis Agent cluster between $W_i \approx 0.17$ and $W_i \approx 0.24$ — with error bars indicating consistent patterns across tasks of the same complexity class.

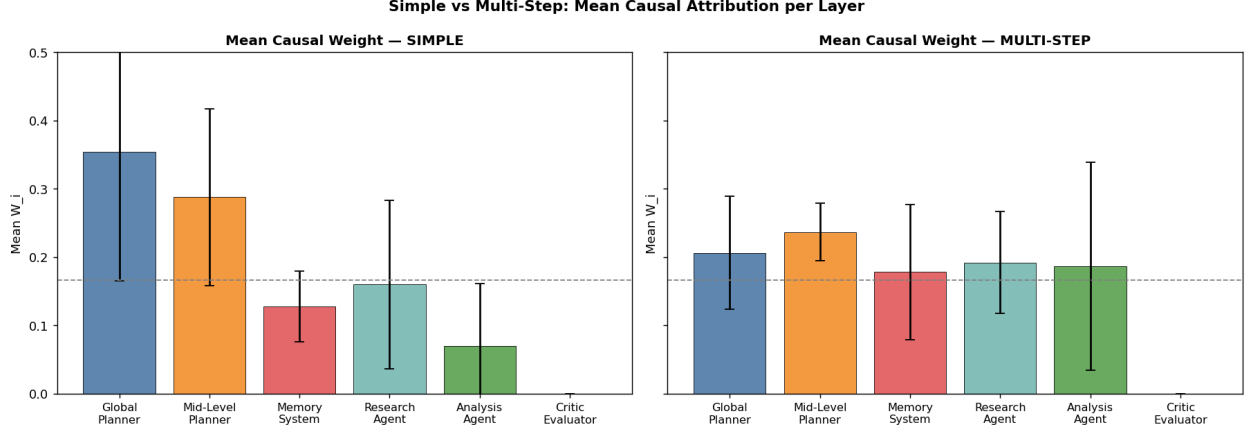


Figure 2: Mean causal weight per layer for simple tasks (left) and multi-step tasks (right), with standard deviation error bars. Simple tasks concentrate causal weight in early planning layers; multi-step tasks distribute it more uniformly across the pipeline. The Critic Evaluator contributes near-zero weight in both conditions.

Finding 3: The dominant causal layer is task-specific. Figure 3 shows the dominant causal layer and its weight for each task. No single layer dominates consistently: Topic Summary is driven by the Research Agent ($W_3 = 0.289$); Factual Q&A by the Mid-Level Planner ($W_1 = 0.421$); Email Drafting by the Global Planner ($W_0 = 0.570$); Research Project Plan by the Mid-Level Planner ($W_1 = 0.298$); System Architecture by the Analysis Agent ($W_4 = 0.411$); Literature Review by the Research Agent ($W_3 = 0.277$); and Product Roadmap by the Global Planner ($W_0 = 0.301$). This variability confirms that causal attribution is task-dependent and cannot be assumed from architecture alone.

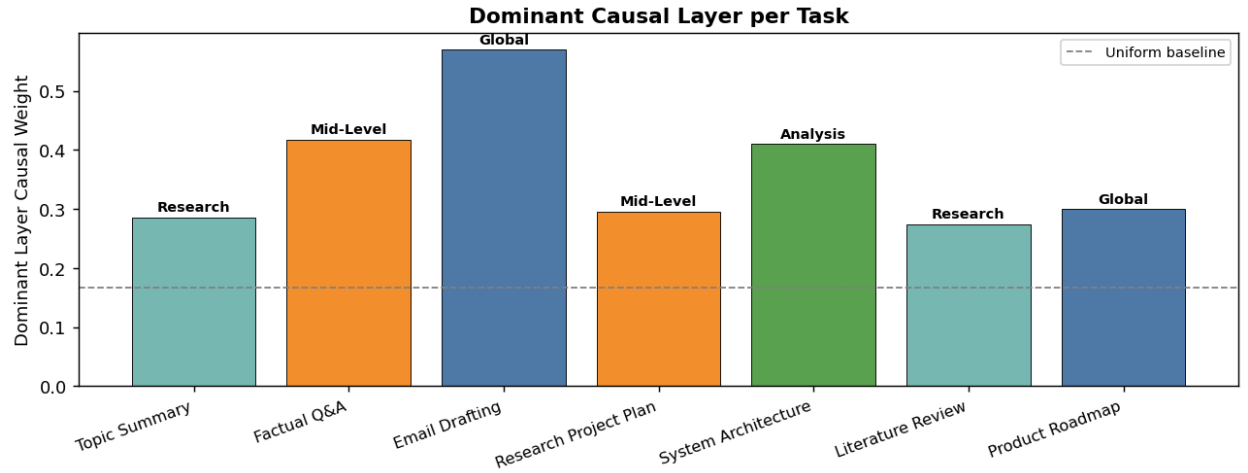


Figure 3: Dominant causal layer and its weight W_i for each of the 7 tasks. Layer labels are annotated above each bar. No single layer dominates across tasks; the dominant layer reflects the specific informational demands of each task type.

Finding 4: Cosine divergence reveals task-level pipeline sensitivity. The Task $\times$ Layer cosine divergence heatmap (Figure 4) reveals the raw causal signal before normalization. Email Drafting produces the largest divergence at the Global Planner ($D_0 = 0.403$) and Mid-Level Planner ($D_1 = 0.203$), indicating high overall pipeline sensitivity to strategic framing. Literature Review and Product Roadmap show uniformly low divergence values, suggesting robustness to single-layer perturbation across all components. The Critic Evaluator column registers $D_5 \approx 0.000$ in all 7 tasks, confirming that perturbing the Critic’s intermediate state produces no measurable change in the final answer.

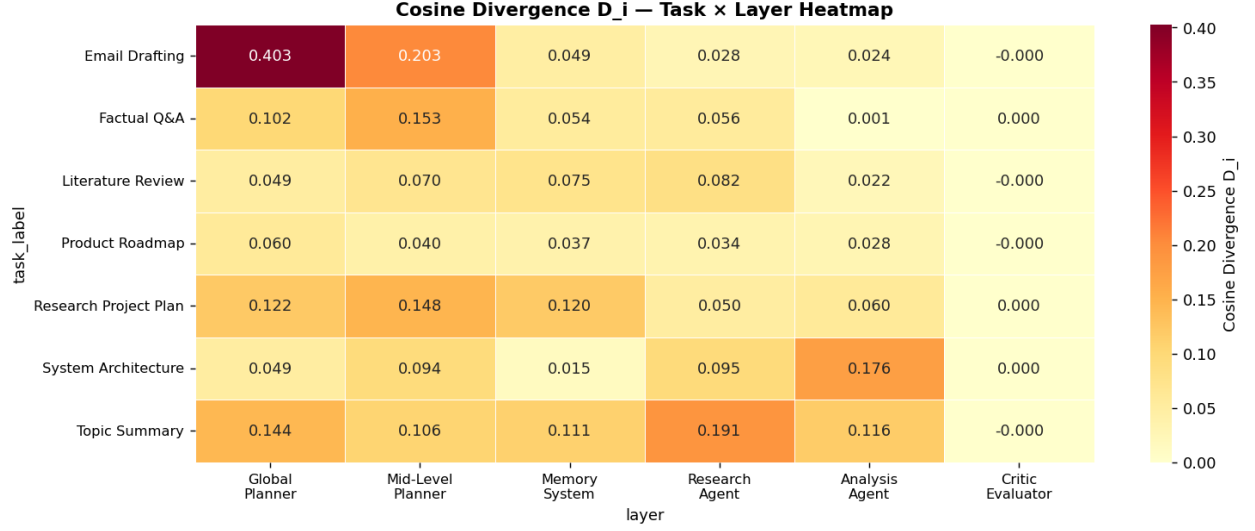


Figure 4: Cosine divergence D_i for each task–layer combination. Cell color encodes divergence magnitude (yellow = low, dark red = high). Email Drafting shows the highest sensitivity to Global Planner perturbation ($D_0 = 0.403$). The Critic Evaluator column is uniformly near-zero across all tasks.

Finding 5: Confabulation rate varies dramatically by layer. Figure 5 shows the proportion of tasks in which each layer’s self-explanation was classified as confabulated. The Critic Evaluator achieves a 100% confabulation rate — in every task, its explanation claimed causal agency over the final output despite having a measured weight of $W_5 \approx 0.000$. The Analysis Agent follows at 43%, and the Global Planner at 29%. Most strikingly, the Research Agent and Mid-Level Planner exhibit 0% confabulation — their self-explanations faithfully reflect their actual causal contribution in every evaluated task.

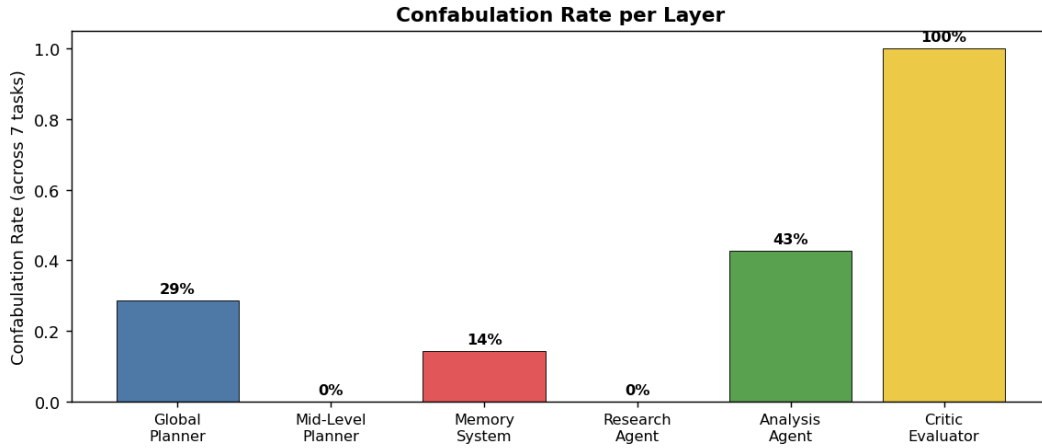


Figure 5: Confabulation rate per layer across all 7 tasks. The Critic Evaluator confabulates in 100% of tasks; the Research Agent and Mid-Level Planner in 0%. Confabulation rates are inversely correlated with layer proximity to concrete, verifiable inputs.

Finding 6: Faithfulness distribution across layers. Figure 6 shows the faithfulness classification distribution per layer (faithful / partial / confabulated) summed across all 7 tasks. The Mid-Level Planner and Memory System show the highest rates of faithful self-explanation (3/7 tasks each). The Research Agent shows only partial and faithful classifications, with zero confabulated cases. The Global Planner and Analysis Agent show mixed profiles, while the Critic Evaluator shows exclusively confabulated classifications across all 7 tasks.

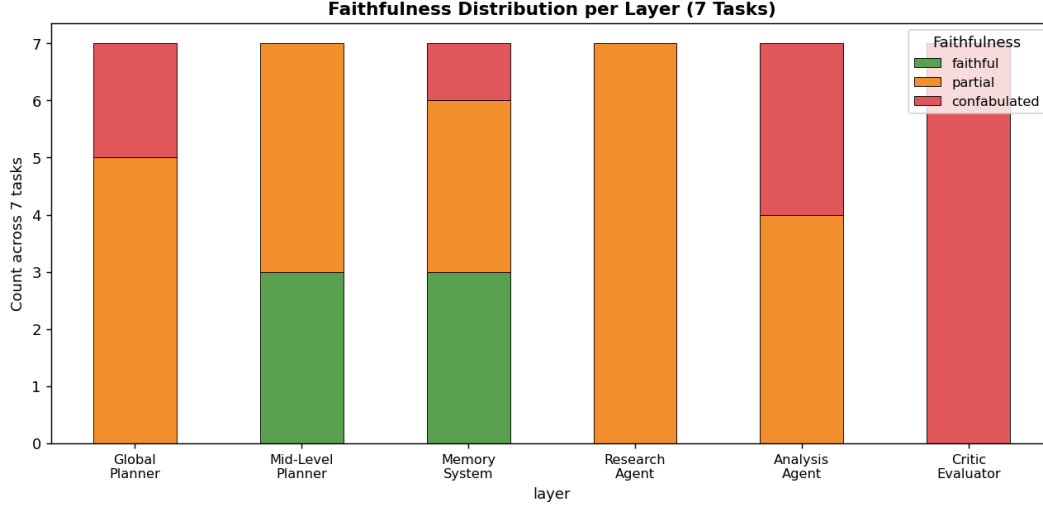


Figure 6: Faithfulness distribution per layer across all 7 tasks. Green bars indicate faithful explanations, orange partial, and red confabulated. The Critic Evaluator is the only layer to show exclusively confabulated explanations across all tasks.

Per-Task HEG Graphs. Figures 7a, 7b, and 7c show representative HEG graphs for Tasks 1, 3, and 5, illustrating the contrast between a simple task with distributed attribution, a simple task with extreme Global Planner dominance, and a multi-step task with downstream layer dominance respectively. In each graph, node size and color intensity encode the causal weight W_i : larger, darker nodes represent greater causal contribution.

HEG — Topic Summary



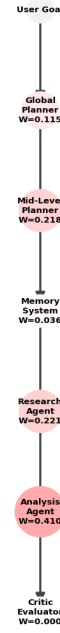
(a) Task 1: Topic Summary. Research Agent dominant ($W_3 = 0.286$). Relatively distributed attribution.

HEG — Email Drafting



(b) Task 3: Email Drafting. Global Planner dominant ($W_0 = 0.570$). Extreme front-loading.

HEG — System Architecture



(c) Task 5: System Architecture. Analysis Agent dominant ($W_4 = 0.410$). Downstream layer drives final output.

Figure 7: HEG graphs for three representative tasks. Node size and color intensity encode causal weight W_i . The Critic Evaluator ($W_5 \approx 0$) appears as a small, pale node at the bottom of each graph.

Total Pipeline Sensitivity. Figure 8 shows the total cosine divergence $\Sigma D = \sum_{i=0}^5 D_i$ per task. Simple tasks generally exhibit higher total sensitivity (Email Drafting: $\Sigma D = 0.707$; Topic Summary: $\Sigma D = 0.668$), while multi-step tasks show lower and more evenly distributed sensitivity (Product Roadmap: $\Sigma D = 0.199$; Literature Review: $\Sigma D = 0.298$). This suggests that multi-step tasks are more robust to single-layer perturbation, likely because their distributed causal structure provides redundancy.

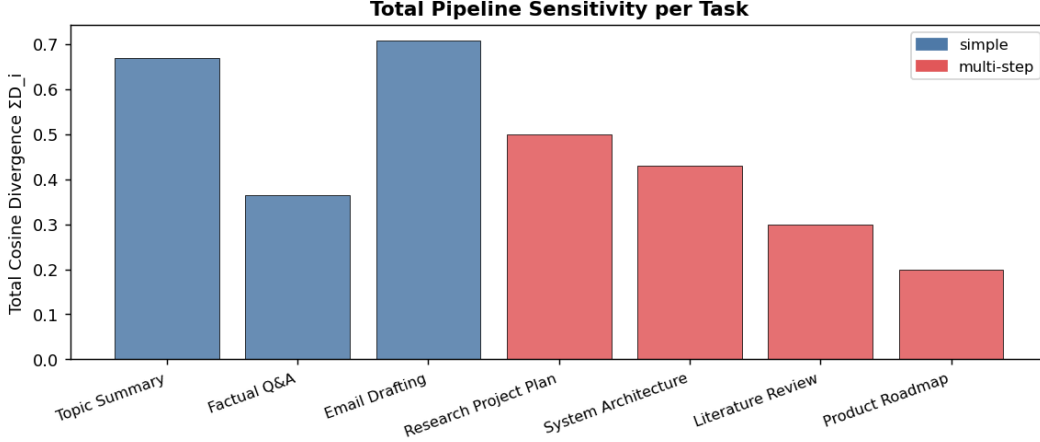


Figure 8: Total cosine divergence ΣD_i per task, indicating overall pipeline sensitivity to perturbation. Simple tasks (blue) show higher total sensitivity than multi-step tasks (red), consistent with their concentrated causal profiles.

Summary Table. Table 1 summarizes the key results for each task, including complexity, dominant causal layer, its weight, and the faithfulness classification breakdown.

Table 1: Summary of HEG results across all 7 tasks. F = Faithful, P = Partial, C = Confabulated (out of 6 layers per task).

Task	Cplx.	Dom. Layer	$W_{\max}$	F	P	C
T1: Topic Summary	Simple	Research Agent	0.286	1	4	1
T2: Factual Q&A	Simple	Mid-Level Plan.	0.418	1	2	3
T3: Email Drafting	Simple	Global Planner	0.570	0	4	2
T4: Res. Proj. Plan	Multi	Mid-Level Plan.	0.296	1	4	1
T5: Sys. Arch.	Multi	Analysis Agent	0.410	1	2	3
T6: Lit. Review	Multi	Research Agent	0.275	1	4	1
T7: Product Roadmap	Multi	Global Planner	0.301	1	3	2

5 Discussion

The Critic Evaluator Paradox. The most striking finding is the simultaneous occurrence of a 100% confabulation rate and a near-zero causal weight ($W_5 \approx 0.000$) in the Critic Evaluator across all 7 tasks. This layer produces the final answer and provides detailed quality scores with explicit justification — yet perturbing its intermediate state causes no measurable change in the pipeline’s final output. The explanation is structural: the Critic’s final answer is determined by the content it receives from upstream layers, not by the scores it assigns. Perturbing the scores changes the Critic’s self-assessment but not the answer it synthesizes from the upstream material. The Critic therefore routinely explains a decision it did not causally make. This mirrors what [Dalla Barba \[1993\]](#) terms fantastic confabulation — the production of elaborate, confident explanations for events the person did not cause — and represents the most severe form of intentional/causal discrepancy the HEG framework can detect.

Front-Loading in Simple Tasks. Simple tasks show a strongly front-loaded causal profile, with the Global Planner and Mid-Level Planner accounting for the majority of causal weight. This reflects the informational structure of simple tasks: once the strategic framing has been established (tone, audience, scope, constraints), downstream layers have little room to alter the final output. The Email Drafting task is the clearest example — the Global Planner’s decision about how to frame the email’s purpose and audience ($W_0 = 0.570$) is so causally dominant that the remaining five layers contribute cumulatively less than 43% of the causal signal. This finding has a practical implication: for simple tasks, investing in prompt quality at the Global Planner level yields disproportionate returns, while adding sophistication to downstream layers provides marginal benefit.

Distributed Causality in Multi-Step Tasks. Multi-step tasks exhibit a substantially more uniform causal distribution, with all layers from Global Planner through Analysis Agent contributing meaningfully. This validates the architectural

justification for hierarchical pipelines: complex tasks genuinely require each specialized layer to perform real causal work. The System Architecture task is illustrative — the Analysis Agent becomes the dominant causal layer ($W_4 = 0.410$), reflecting that for a task requiring architectural decision-making, the interpretation and synthesis of collected information is the primary causal driver, not the strategic framing alone. The HEG framework makes this empirically visible rather than architecturally assumed.

Execution Layers as Faithful Self-Reporters. The zero confabulation rate of the Research Agent and Mid-Level Planner is a theoretically meaningful finding. Both layers interact with concrete, verifiable inputs — subtask specifications and structured data sources — and produce structured outputs with clear correspondence to their inputs. This reduces the gap between what the layer “knows” about its own process and what actually drove its output. Abstract evaluation layers (Critic, Analysis Agent) operate over already-processed, high-dimensional content where the mapping from input to output is less transparent even to the layer itself, creating conditions for post-hoc rationalization. This aligns with Johnson [1991]’s source monitoring framework: agents with concrete external input markers self-report more accurately than those performing abstract internal inference.

Implications for Agent Design. The HEG findings suggest two practical design principles. First, explanation requests should be directed primarily at layers with low confabulation rates (Research Agent, Mid-Level Planner) rather than at the final output layer, which exhibits the highest confabulation rate. Second, for systems where downstream layers have high confabulation rates, causal weight monitoring via periodic perturbation experiments can serve as a runtime faithfulness check — flagging when a layer’s self-report diverges from its measured causal contribution.

Limitations. The current evaluation has several limitations that future work should address. First, all pipeline layers and the Perturbation Engine use the same backbone model (GPT-4o); cross-model confabulation patterns may differ. Second, the minimal perturbation constraint is enforced via prompting rather than through a formal perturbation budget, which limits the strict comparability of divergence magnitudes across layers. Third, cosine divergence captures semantic shift in embedding space but may not detect structural changes in output format or reasoning chain that are semantically neutral. Fourth, the 7-task evaluation covers general-purpose domains; specialized domains (medical, legal, scientific) may exhibit different causal patterns and should be evaluated separately.

6 Conclusion

I introduced the Hierarchical Explanation Graph (HEG), a two-level causal explainability framework for hierarchical LLM-based multi-agent systems. The framework combines simultaneous intentional explanation capture at each pipeline layer with a systematic program of layer-targeted perturbation experiments, producing experimentally grounded causal weights that quantify the true causal contribution of each layer to the final pipeline output.

Across 7 tasks and 42 perturbation experiments, the HEG framework reveals systematic intentional/causal discrepancy that would be invisible to self-report-based explainability methods. The Critic Evaluator — the layer that produces and explicitly justifies the final answer — exhibits a 100% confabulation rate and near-zero causal weight, demonstrating that the apparent locus of decision-making is causally decoupled from the true locus. Execution-level layers with concrete, verifiable inputs (Research Agent, Mid-Level Planner) exhibit zero confabulation, suggesting that faithful self-explanation is achievable when layer inputs are structured and the input-output mapping is tractable.

These findings establish that causal explainability in hierarchical agents is not achievable through self-report analysis alone, and that the cognitive science framework of confabulation provides both a theoretical grounding and a practical vocabulary for characterizing the failure modes of LLM self-explanation. Future work will extend the HEG framework to cross-model evaluation, formal perturbation budgets, integration with token-level mechanistic interpretability, and real-time faithfulness monitoring in deployed agent systems.

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